Analytic Geometry through Normals, Line Pencils, and Quadratic Forms

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§1.1 A line is an affine level set, not a collection of unrelated formulas

A line in the plane can be written

$$H = \{x \in \mathbb{R}^2 : n \cdot x = c\}, \qquad n \neq 0.$$

The vector $n = (a, b)$ is normal to the line, and a direction vector is

$$d = (-b, a).$$

If $p \in H$, every point of the line has the parametric form

$$x = p + td, \qquad t \in \mathbb{R}.$$

Conversely, eliminating t gives

$$a(x_1 - p_1) + b(x_2 - p_2) = 0.$$

This single description includes vertical lines, where slope notation fails.

The familiar equations are coordinate choices for the same object. If $b \neq 0$,

$$ax + by = c \quad \iff \quad y = -\frac{a}{b}x + \frac{c}{b}.$$

If a line passes through p_1, p_2, its direction is $p_2 - p_1$. If it meets the coordinate axes away from the origin, dividing by its intercepts produces

$$\frac{x}{a} + \frac{y}{b} = 1.$$

None of these forms is universally best; the normal form survives every orientation, while slope form is efficient only in the nonvertical chart.

In homogeneous coordinates, the line is represented by the coefficient triple

$$\ell = [a : b : -c],$$

and incidence with a projective point $X = [x : y : z]$ is the bilinear equation

$$ax + by - cz = 0.$$

The homogeneous form will later make points at infinity and polar lines ordinary parts of the calculation.

§1.2 Angles and Euclidean distance come from the normal vector

Two lines with normals n_1, n_2 are parallel exactly when the normals are proportional. They are perpendicular exactly when

$$n_1 \cdot n_2 = 0,$$

because the direction of each line is a quarter-turn of its normal.

For

$$H = \{x : n \cdot x = c\}$$

and a point x_0, the signed displacement in the normal direction is

$$\frac{n \cdot x_0 - c}{\|n\|_2}.$$

Hence

$$\boxed{\operatorname{dist}_2(x_0, H) = \frac{|n \cdot x_0 - c|}{\|n\|_2}.}$$

To derive it, let

$$p = x_0 - \frac{n \cdot x_0 - c}{\|n\|_2^2} n.$$

Then $n \cdot p = c$, so $p \in H$, and $x_0 - p$ is normal to the line. Any other point of H differs from p by a tangent vector, so the Pythagorean theorem makes p the closest point.

For parallel lines

$$n \cdot x = c_1, \qquad n \cdot x = c_2,$$

the same formula gives

$$\operatorname{dist}_2(H_1, H_2) = \frac{|c_1 - c_2|}{\|n\|_2}.$$

The denominator is not a memorized normalization constant; it compensates for rescaling the defining equation.

§1.3 Distance to a hyperplane is governed by the dual norm

The Euclidean formula has a normed-space version. Let $\|\cdot\|$ be any norm on $\mathbb{R}^n$, and let φ be a nonzero linear functional. Its dual norm is

$$\|\varphi\|_* = \sup_{\|u\| \le 1} |\varphi(u)|.$$

For the hyperplane

$$H = \{x : \varphi(x) = c\},$$

one has

$$\boxed{\operatorname{dist}(x_0, H) = \frac{|\varphi(x_0) - c|}{\|\varphi\|_*}.}$$

Indeed, for every $x \in H$,

$$|\varphi(x_0) - c| = |\varphi(x_0 - x)| \le \|\varphi\|_* \|x_0 - x\|,$$

which gives the lower bound. In finite dimensions, the supremum defining the dual norm is attained by a unit vector u; moving from x_0 in the signed u-direction reaches equality.

For $\varphi(x, y) = ax + by$, the dual of ℓ^1 is ℓ^∞, so

$$\operatorname{dist}_1((x_0, y_0), H) = \frac{|ax_0 + by_0 - c|}{\max(|a|, |b|)}.$$

The dual of ℓ^∞ is ℓ^1, giving

$$\operatorname{dist}_\infty((x_0, y_0), H) = \frac{|ax_0 + by_0 - c|}{|a| + |b|}.$$

The familiar square root in the Euclidean formula is therefore a special case of dual-norm geometry.

§1.4 Circles are quadratic level sets, and subtraction reveals their common geometry

A circle with center $m = (a, b)$ and radius r is

$$\|x - m\|_2^2 = r^2.$$

Expanding gives

$$x^2 + y^2 + Dx + Ey + F = 0,$$

while completing squares recovers

$$m = \left(-\frac{D}{2}, -\frac{E}{2}\right),$$

$$r^2 = \frac{D^2 + E^2 - 4F}{4}.$$

The sign of this final quantity distinguishes a real circle, a point circle, and no real locus.

For a point p, define its power with respect to the circle by

$$\operatorname{Pow}_C(p) = \|p - m\|^2 - r^2.$$

If a line through p meets the circle at a, b, then

$$\operatorname{Pow}_C(p) = \overline{pa}\,\overline{pb}$$

using directed lengths. This follows by substituting the line parametrization $p + td$ into the circle equation: the product of the two parameter roots is the constant term divided by the leading coefficient.

For two circles, the locus of equal powers is their radical axis. Subtracting

$$C_1 : x^2 + y^2 - 4x = 0$$

and

$$C_2 : x^2 + y^2 - 2y - 3 = 0$$

cancels the quadratic terms and gives the line

$$-4x + 2y + 3 = 0.$$

The radical axis is linear because it compares two quadratic functions with the same leading form.

§1.5 A fractional range problem is a pencil of lines through one point

Consider

$$\mu = \frac{x}{y-3}$$

under the constraint

$$|x| + |y| \le 1.$$

For fixed μ, rewrite the equation as

$$x = \mu(y-3).$$

This is a line through

$$P = (0,3).$$

The range problem asks which lines in the pencil through P meet the diamond

$$K = \{(x,y) : |x| + |y| \le 1\}.$$

Because K is convex and $P \notin K$, the extreme parameters occur at supporting lines from P to K.

The support function of the ℓ^1-unit ball is

$$h_K(n) = \max_{x \in K} n \cdot x = \|n\|_\infty.$$

The line can be written

$$(1, -\mu) \cdot (x,y) = -3\mu.$$

It meets K precisely when

$$|3\mu| \le h_K(1, -\mu) = \max(1, |\mu|).$$

If $|\mu| > 1$, this is impossible. If $|\mu| \le 1$, it becomes

$$3|\mu| \le 1.$$

Therefore

$$\boxed{\mu \in \left[-\frac{1}{3}, \frac{1}{3}\right].}$$

The endpoints correspond to the supporting lines through the vertices $(1,0)$ and $(-1,0)$. Convex duality has not replaced the elementary picture; it has explained why the endpoints are support data.

§1.6 Ratios on a graph are radial coordinates

For a point $(x, f(x))$ with $x \ne 0$, the ratio

$$\frac{f(x)}{x}$$

is the slope of the ray from the origin to the graph point. Counting solutions of

$$\frac{f(x)}{x} = m$$

is therefore the same as counting intersections between the graph and the line

$$y = mx.$$

A rational expression has become a radial-projection problem.

This viewpoint also identifies tangency thresholds. As m varies, the number of intersections can change only when the line becomes tangent to the graph or passes through a singular point. Algebraically, solving

$$f(x) = mx$$

together with

$$f'(x) = m$$

detects those critical slopes. The graphical and derivative descriptions are the same incidence event viewed globally and locally.

§1.7 A perpendicular moving-line problem is a circle problem

Let fixed points be

$$A = (-3, 0), \qquad B = (1, 6).$$

Suppose lines through A and B meet at P and are perpendicular. Then

$$\angle APB = 90^\circ,$$

so Thales' theorem places P on the circle with diameter AB.

The right triangle gives

$$PA^2 + PB^2 = AB^2.$$

Hence

$$PA\, PB \le \frac{PA^2 + PB^2}{2} = \frac{AB^2}{2}.$$

Since

$$AB^2 = (1 + 3)^2 + 6^2 = 52,$$

one obtains

$$\boxed{PA\, PB \le 26.}$$

Equality occurs when $PA = PB$, namely at the intersections of the diameter circle with the perpendicular bisector of AB.

The coordinate version would introduce two slopes and a quadratic constraint. The circle formulation identifies the locus first, after which the optimization is a one-line inequality.

§1.8 A general conic is an affine quadratic form

Write an affine conic as

$$F(x) = x^T Q x + 2\ell^T x + f = 0, \qquad Q = Q^T.$$

An orthogonal change of coordinates diagonalizes Q, removing the cross term without changing Euclidean angles. If Q is invertible, translating by

$$x = u - Q^{-1}\ell$$

removes the linear term. Rotation handles directions; translation handles the center.

Consider

$$xy - x - y = 0.$$

Set

$$u = \frac{x+y}{\sqrt{2}}, \qquad v = \frac{x-y}{\sqrt{2}}.$$

Then

$$xy = \frac{u^2 - v^2}{2}, \qquad x + y = \sqrt{2}\,u.$$

The equation becomes

$$u^2 - v^2 - 2\sqrt{2}\,u = 0,$$

or

$$\boxed{(u - \sqrt{2})^2 - v^2 = 2.}$$

The original rectangular-looking equation is a hyperbola whose principal axes are the eigenvectors of the quadratic matrix

$$Q = \begin{pmatrix} 0 & 1/2 \\ 1/2 & 0 \end{pmatrix}.$$

§1.9 Rank and the line at infinity distinguish conic types

Homogenizing a conic gives

$$X^T A X = 0, \qquad X = (x, y, z)^T,$$

for a symmetric 3×3 matrix A. The conic is degenerate when A loses rank. For example,

$$xy = 0$$

is the union of two lines and has a rank-deficient homogeneous matrix.

The intersection with the line at infinity $z = 0$ records affine directions.

- An ellipse
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = z^2$$
has no real point at infinity.
- A parabola
$$y^2 = 2pxz$$
meets the line at infinity at one double point $[1 : 0 : 0]$.
- A hyperbola
$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = z^2$$
has two real points at infinity, corresponding to its asymptotic directions.

For

$$xy - x - y = 0,$$

the projective closure is

$$XY - XZ - YZ = 0.$$

At infinity,

$$Z = 0 \quad \Longrightarrow \quad XY = 0,$$

so the two points are

$$[1 : 0 : 0], \qquad [0 : 1 : 0].$$

They are the projective endpoints of the two asymptote directions revealed by the standard hyperbola form.

§1.10 Polarity turns the conic matrix into its tangent geometry

Let a nondegenerate projective conic be

$$F(X) = X^T AX = 0, \qquad A = A^T.$$

For a point p, its polar line is

$$p^T AX = 0.$$

If p lies on the conic, this polar is the tangent line. Indeed, the differential is

$$dF_p(H) = 2p^T AH.$$

The tangent condition at p is

$$p^T A(X - p) = 0.$$

Since $p^T Ap = 0$, this reduces to

$$p^T AX = 0.$$

For the conic

$$XY - XZ - YZ = 0,$$

use the symmetric matrix

$$A = \begin{pmatrix} 0 & 1 & -1 \\ 1 & 0 & -1 \\ -1 & -1 & 0 \end{pmatrix},$$

which represents twice the equation. The affine point $(2, 2)$ corresponds to

$$p = [2 : 2 : 1]$$

and lies on the conic. Since

$$Ap = \begin{pmatrix} 1 \\ 1 \\ -4 \end{pmatrix},$$

its polar, hence its tangent, is

$$X + Y - 4Z = 0.$$

In the affine chart $Z = 1$,

$$\boxed{x + y = 4.}$$

The same answer follows from the gradient of $xy-x-y$ at $(2, 2)$, but the polar calculation remains valid in homogeneous coordinates and transforms naturally with the conic under projective changes of coordinates.